2. Write the python program to solve 8-Queen problem

Aim

To develop a Python program to solve the **8-Queen Problem** using the **Backtracking Algorithm** by accepting the **board size (N)** from the user and placing **N queens** on an **N × N chessboard** such that no two queens attack each other.

Algorithm

1. Start the program.
2. Read the number of queens (N) from the user.
3. Create an empty N × N chessboard.
4. Begin placing queens column by column.
5. For each row in the current column:
    - Check whether placing the queen is safe.
    - If safe, place the queen.
    - Recursively place queens in the next column.
    - If a solution is found, stop.
    - Otherwise, remove the queen (Backtrack).
6. Repeat until all queens are placed.
7. Display the chessboard if a solution exists; otherwise display "No Solution Exists."
8. Stop.

Program

```
# Function to check whether a queen can be placed safely
def is_safe(board, row, col, n):

    # Check left side of current row
    for i in range(col):
        if board[row][i] == 1:
            return False

    # Check upper-left diagonal
    i = row
    j = col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    # Check lower-left diagonal
    i = row
    j = col
    while i < n and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
```

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        j -= 1

    return True


# Backtracking function
def solve_nqueen(board, col, n):

    if col >= n:
        return True

    for i in range(n):
        if is_safe(board, i, col, n):

            board[i][col] = 1

            if solve_nqueen(board, col + 1, n):
                return True

            board[i][col] = 0

    return False


# Main Program
n = int(input("Enter the number of Queens (N): "))

board = [[0 for i in range(n)] for j in range(n)]

if solve_nqueen(board, 0, n):

    print("\nSolution Exists:\n")

    for row in board:
        for cell in row:
            if cell == 1:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()

else:
    print("No Solution Exists.")
```

Input

```
Enter the number of Queens (N): 8
```

Output

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Result

The Python program was successfully implemented to solve the **8-Queen Problem** using the **Backtracking Algorithm**. The program accepts the **number of queens (N)** from the user, places the queens on an **N × N chessboard** such that no two queens attack each other, and displays one valid solution if it exists. This demonstrates the effectiveness of the backtracking technique in solving constraint satisfaction problems.